Example 9.17

Calculate $\lim_{t\to 1}\frac{\sqrt{t}-1}{t-1}$.

Solution

$$\lim_{x\to 1}\frac{\sqrt{t}-1}{t-1} = \lim_{x\to 1}\frac{t^{\frac{1}{2}}-1^{\frac{1}{2}}}{t-1} = \frac{1}{2}(1)^{\frac{1}{2}-1} = \frac{1}{2}.$$

Example 9.18

Find $\lim_{x\to 0}\frac{(2+x)^5-2^5}{x}$.

Solution

Put $2 + x = y$ so that as $y \to 2$ as $x \to 0$.

Therefore, $\lim_{x\to 0}\frac{(2+x)^5-2^5}{x} = \lim_{y\to 2}\frac{y^5-2^5}{y-2} = 5(2^4) = 80$.

Example 9.19

Find the positive integer n so that $\lim_{x\to 3}\frac{x^n-3^n}{x-3} = 27$.

Solution

$\lim_{x\to 3}\frac{x^n-3^n}{x-3} = n.3^{n-1} = 27$

That is $n.3^{n-1} = 3\times 3^2 = 3\times 3^{3-1} \Rightarrow n = 3$.

Example 9.20

Find the relation between a and b if $\lim_{x\to 3} f(x)$ exists where $f(x) = \begin{cases} ax+b & \text{if } x>3 \\ 3ax-4b+1 & \text{if } x<3 \end{cases}$.

Solution

$\lim_{x\to 3^-} f(x) = 9a - 4b + 1$

$\lim_{x\to 3^+} f(x) = 3a + b$. Now the existence of limit forces us to have

$\lim_{x\to 3^-} f(x) = \lim_{x\to 3^+} f(x)$.

$\Rightarrow 9a - 4b + 1 = 3a + b$

$\Rightarrow 6a - 5b + 1 = 0.$

EXERCISE 9.2

Evaluate the following limits :

(1) $\lim_{x\to 2}\frac{x^4-16}{x-2}$

(2) $\lim_{x\to 1}\frac{x^m-1}{x^n-1}$, m and n are integers.

(3) $\lim_{\sqrt{x}\to 3}\frac{x^2-81}{\sqrt{x}-3}$

(4) $\lim_{h\to 0}\frac{\sqrt{x+h}-\sqrt{x}}{h}, x>0$

(5) $\lim_{x\to 5}\frac{\sqrt{x+4}-3}{x-5}$

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